

GitHub API Field Guide

The GitHub API stops working quietly: a list that returns thirty items, a webhook failing for a month, a 403 with no Retry-After. Every script here is read only.

40 fixes, each one a written note with diagrams and a small Python and Node.js script that finds the problem and repairs it. All 40 have a script in the public repo.

Before you run anything. Every script starts in a dry run: it reports what it would change and writes nothing. Read that output first, then set `DRY_RUN=false` when you are satisfied it has understood your data.

Scripts: github.com/allanninal/github-api-fixes

Guides: allanninal.dev/github

All 14 repos: github.com/allanninal

The fixes

1 a permission error is disguised as 404 Not Found

The repository is open in a browser tab in front of you. The script asks for the same repository and gets 404 {"message":"Not Found"}. Somebody checks the spelling, then the owner name, then the case of both, then writes a ticket saying the API is broken. The API is not broken. It is refusing to tell you that the repository exists, because telling you would leak the existence of a private repository to a credential that has no business knowing about it.

[github_404_triage.py](#)

<https://www.allanninal.dev/github/404-masking-403/>

<https://github.com/allanninal/github-api-fixes/tree/main/404-masking-403>

2 GITHUB_TOKEN gets 1,000 an hour, shared across the repo

The script works on your laptop. It makes about 1,400 calls, finishes in two minutes, and nobody thinks about it again until it is moved into a workflow. There it gets to roughly call 1,000 and starts collecting 403. Same code, same repository, same endpoints. The only thing that changed is which credential is in the environment, and that credential was handed a ceiling a fifth the size of the one you tested against.

[github_actions_token_budget.py](#)

<https://www.allanninal.dev/github/actions-token-repo-scoped-limit/>

<https://github.com/allanninal/github-api-fixes/tree/main/actions-token-repo-scoped-limit>

3 resource not accessible by integration on one endpoint

Nineteen endpoints work. The twentieth returns 403 {"message":"Resource not accessible by integration"}, which names no permission, no resource and no level, and reads like a platform bug rather than a configuration one. It is the one failure in this cluster where GitHub actually tells you the answer: it is in the x-accepted-github-permissions response header on that very 403, which almost no HTTP client shows you by default.

[github_app_permission_diff.py](#)

<https://www.allanninal.dev/github/app-permission-missing/>

<https://github.com/allanninal/github-api-fixes/tree/main/app-permission-missing>

4 401 Bad credentials on every endpoint, even public ones

401 {"message":"Bad credentials"} on every endpoint you try, including ones that need no credential at all. The token is right there in the environment, it was working yesterday, and the error names nothing: not the account, not the scope, not which of the six things that produce this message actually happened.

[github_401_provenance.py](#)

<https://www.allanninal.dev/github/bad-credentials-401/>

<https://github.com/allanninal/github-api-fixes/tree/main/bad-credentials-401>

5 the client still sends a username and password to the API

401 {"message": "Support for password authentication was removed. Please use a personal access token instead."}. The reflex is to go and mint a token, which is right, and then to paste it where the password was, which is half right, and then to be surprised when a different call still fails. The credential was never the problem. The envelope was.

[github_auth_scheme_check.py](#)

<https://www.allanninal.dev/github/basic-auth-password-removed/>

<https://github.com/allanninal/github-api-fixes/tree/main/basic-auth-password-removed>

6 a classic PAT passed its expiry and everything broke at once

The integration ran for eleven months without anyone touching it. At 09:14 on a Tuesday every call started returning 401 Bad credentials, all at once, on every endpoint, with no deploy, no repository change and no org announcement. A classic personal access token reached its expiry date. So did a great many other theories that morning, and the API will not tell you which one was right.

[github_credential_differential.py](#)

<https://www.allanninal.dev/github/classic-pat-expired/>

<https://github.com/allanninal/github-api-fixes/tree/main/classic-pat-expired>

7 code search is billed to its own 10 a minute bucket

The script walks the org, calling GET /search/code once per repository. It gets through nine of them. The tenth returns 403, and GET /rate_limit says you have 4,987 requests left in the hour. Both statements are correct, because the request that was refused was never being counted in the bucket you just looked at.

[github_code_search_budget.py](#)

<https://www.allanninal.dev/github/code-search-bucket-exhausted/>

<https://github.com/allanninal/github-api-fixes/tree/main/code-search-bucket-exhausted>

8 the compare endpoint stops at 250 commits and says nothing

The release-notes generator diffs v4.2.0...v4.3.0 and produces a changelog that looks entirely plausible. It has 250 entries. The release contains 812 commits, and the ones it dropped are not the boring ones at the end — the shape of what came back is not what the code assumed at all.

[github_compare_truncation.py](#)

<https://www.allanninal.dev/github/compare-250-commit-cap/>

<https://github.com/allanninal/github-api-fixes/tree/main/compare-250-commit-cap>

9 the same webhook URL is registered on the org and the repo

The bot comments twice on every pull request. Someone checks the receiver for a retry loop, finds none, and blames GitHub for sending duplicates. GitHub is sending exactly one copy of the event to each hook that asked for it — and two hooks asked, one on the repository and one on the organization, both pointing at the same URL.

[github_duplicate_hooks.py](#)

<https://www.allanninal.dev/github/duplicate-webhooks/>

<https://github.com/allanninal/github-api-fixes/tree/main/duplicate-webhooks>

10 rotating the token invalidates every cached ETag at once

The graph is a sawtooth and it is a very tidy one. Quota consumption sits near zero for fifty-odd minutes, jumps, and settles back, every hour, on the hour. Nothing in the schedule matches. The poller runs every thirty seconds and has done for months. What runs hourly is the installation token.

[github_etag_credential_check.py](#)

<https://www.allanninal.dev/github/etag-invalidated-by-token-rotation/>

<https://github.com/allanninal/github-api-fixes/tree/main/etag-invalidated-by-token-rotation>

11 the installation covers only some repositories, silently

The scanner reports clean. Every repository it looked at was fine, every check passed, and the summary at the bottom says so. It looked at twelve repositories. The organization has a hundred and forty. Nothing errored, nothing warned, and no line of that report is untrue — the App installation was set to selected repositories at some point in 2023 and the other hundred and twenty-eight have never been inside its field of view.

[github_app_coverage_audit.py](#)

<https://www.allanninal.dev/github/installation-repository-selection-partial/>

<https://github.com/allanninal/github-api-fixes/tree/main/installation-repository-selection-partial>

12 some endpoints refuse an installation token whatever it holds

The App is installed. The JWT signs, the installation token mints, and nineteen calls in a row return exactly what they should. Then GET /user comes back 403 {"message": "Resource not accessible by integration"}, somebody adds a permission, every installer is emailed to accept the upgrade, and the same 403 arrives again the next morning.

[github_endpoint_audience.py](#)

<https://www.allanninal.dev/github/installation-token-rejected-by-endpoint/>

<https://github.com/allanninal/github-api-fixes/tree/main/installation-token-rejected-by-endpoint>

13 a GitHub App JWT that expires in an hour is refused

The private key is right, the App exists, the signature verifies and the request still comes back 401 {"message": "'Expiration time' claim ('exp') is too far in the future"}. Three people regenerate the key. Somebody rewrites the signing code in a different library. The defect is one number, chosen three lines above the request, and it was wrong before anything was sent.

[github_app_jwt_claims.py](#)

<https://www.allanninal.dev/github/jwt-exp-too-far-future/>

<https://github.com/allanninal/github-api-fixes/tree/main/jwt-exp-too-far-future>

14 only the first page is read because the Link header is ignored

The request returned 200. The JSON is a well-formed array of pull requests. Your audit read it, counted 30, and reported that the repository is tidy. There are 340 open pull requests. Nothing failed, nothing was logged, and the number you are now acting on is wrong by an order of magnitude — the answer was complete for one page and the rest was advertised in a header nobody read.

[github_link_header_audit.py](#)

<https://www.allanninal.dev/github/link-header-not-followed/>

<https://github.com/allanninal/github-api-fixes/tree/main/link-header-not-followed>

15 the endpoint accepts a scope your token was never given

Nineteen calls work. The twentieth returns 403 {"message": "Must have admin rights to Repository."}, or worse, a bare 404 on a repository that is open in a browser tab beside you. The token is valid, it is not expired, it has not been revoked, and it is missing exactly one word from the list it was created with.

[github_scope_diff.py](#)

<https://www.allanninal.dev/github/missing-oauth-scope/>

<https://github.com/allanninal/github-api-fixes/tree/main/missing-oauth-scope>

16 polling without ETags spends full quota on unchanged data

The dashboard polls eight endpoints every thirty seconds and burns through 5,000 requests before lunch. Almost nothing it fetches has changed. GitHub has been sending an etag on every one of those responses, and every response that comes back 304 Not Modified is free — it does not count against the rate limit at all. This is the one problem in this section where the fix pays for itself in a number you can print.

[github_etag_saving.py](#)

<https://www.allanninal.dev/github/no-conditional-requests/>

<https://github.com/allanninal/github-api-fixes/tree/main/no-conditional-requests>

17 one user revoked your app and only their token is dead

One customer's sync has been failing since Thursday. Everybody else is fine. The token has not expired, because OAuth user tokens do not expire; nothing was deployed; the error is 401 Bad credentials and the retry loop has been dutifully re-attempting it every fifteen minutes for four days. The user clicked Revoke on a settings page you will never see.

[github_user_token_liveness.py](#)

<https://www.allanninal.dev/github/oauth-token-revoked-by-user/>

<https://github.com/allanninal/github-api-fixes/tree/main/oauth-token-revoked-by-user>

18 **a read-only job holds a token that can delete repositories**

There is no incident to attach this to. The job runs every ten minutes, lists open pull requests, posts a count to a dashboard and has not failed once this year. The token it uses was created in four seconds by ticking repo, and it can force-push to every private repository in the organization, rewrite webhooks and delete the lot.

[github_scope_blast_radius.py](#)

<https://www.allanninal.dev/github/over-scoped-token/>

<https://github.com/allanninal/github-api-fixes/tree/main/over-scoped-token>

19 **per_page is unset so every list costs 3.3x more requests**

The job is correct. It follows rel="next" to the end, it reads every issue, and it burns through the hourly quota by lunchtime. Nothing is broken and nothing needs debugging — it is simply making three and a third times as many requests as it needs to, because per_page was never set and the default is 30.

[github_per_page_audit.py](#)

<https://www.allanninal.dev/github/per-page-default-30/>

<https://github.com/allanninal/github-api-fixes/tree/main/per-page-default-30>

20 **the x-poll-interval header is ignored on events endpoints**

The events consumer polls every five seconds because five seconds felt responsive. It gets the same page back seven hundred times an hour. On every one of those responses GitHub has been returning a header that says how long to wait before the next one, and the client has never read it.

[github_poll_interval_check.py](#)

<https://www.allanninal.dev/github/poll-interval-header-ignored/>

<https://github.com/allanninal/github-api-fixes/tree/main/poll-interval-header-ignored>

21 **the integration polls for events a webhook would push**

The integration works. It notices new pull requests, it picks up comments, it has never lost anything. It also makes 4,320 requests an hour to do it, notices each of those events an average of thirty seconds after it happened, and would notice nothing at all if the poll ran once a day instead. There is no bug here. There is a design that was never revisited.

[github_webhook_vs_poll.py](#)

<https://www.allanninal.dev/github/polling-instead-of-webhooks/>

<https://github.com/allanninal/github-api-fixes/tree/main/polling-instead-of-webhooks>

22 **core REST quota is exhausted and every call returns 403**

Every endpoint fails at once, with the same status, in the same second. That pattern says outage or bad credentials, and it is neither: the hourly bucket is empty, and an empty bucket refuses everything equally. The awkward part is that by the time you are reading the 403 the interesting question has already gone past. Not are we out, but at what rate did we spend it, and would that rate have fitted.

[github_quota_forecast.py](#)

<https://www.allanninal.dev/github/rate-limit-core-exhausted/>

<https://github.com/allanninal/github-api-fixes/tree/main/rate-limit-core-exhausted>

23 requests go out anonymous and are capped at 60 an hour

It works on the first run and dies on the fourth. On a laptop it survives a couple of minutes; in CI, behind a shared NAT address, it is refused almost immediately and the run before yours gets the blame. Nothing in the code is wrong. The token simply is not arriving, and GitHub does not treat that as an error. It serves you anyway, as a stranger, from a bucket sixty deep.

[github_auth_tier_check.py](#)

<https://www.allanninal.dev/github/rate-limit-unauthenticated/>

<https://github.com/allanninal/github-api-fixes/tree/main/rate-limit-unauthenticated>

24 the client ignores retry-after and keeps hammering the API

The log shows four hundred consecutive 403s, one second apart, for eleven minutes. The retry logic is working perfectly: it catches the error, waits, tries again, never gives up. GitHub said retry-after: 120 on the very first one, and the client threw that header away along with the rest of the response.

[github_backoff_plan.py](#)

<https://www.allanninal.dev/github/retry-after-ignored/>

<https://github.com/allanninal/github-api-fixes/tree/main/retry-after-ignored>

25 org lists silently omit SSO-enforced organizations

The user belongs to six organizations. GET /user/orgs returns four. Status 200, valid JSON, no errors array, nothing in the body that hints anything is absent. The two organizations that enforce SAML single sign-on for a token that was never authorized against them are simply not in the list, and the only place GitHub mentions it is a response header called X-GitHub-SSO.

[github_sso_partial_results.py](#)

<https://www.allanninal.dev/github/saml-partial-results/>

<https://github.com/allanninal/github-api-fixes/tree/main/saml-partial-results>

26 search returns at most 1,000 results whatever total_count says

The response says "total_count": 24831. You page through it at 100 per request, and on page 11 the API returns 422 Validation Failed with "Only the first 1000 search results are available". The count was true. The results were never there to fetch — total_count describes the match set, not the part of it you are allowed to page through.

[github_search_cap_audit.py](#)

<https://www.allanninal.dev/github/search-1000-result-cap/>

<https://github.com/allanninal/github-api-fixes/tree/main/search-1000-result-cap>

27 search has its own 30-per-minute bucket and drains separately

The job loops over four hundred repositories and runs one search in each. Around the thirtieth it starts returning 403, and the part that makes no sense is that everything else keeps working: the repository reads in the same loop, on the same token, in the same second, are fine. Two buckets, and only one of them is empty. The error message does not mention which.

[github_search_budget.py](#)

<https://www.allanninal.dev/github/search-bucket-exhausted/>

<https://github.com/allanninal/github-api-fixes/tree/main/search-bucket-exhausted>

28 **over 100 concurrent requests trips a secondary rate limit**

Someone replaces a for loop with a Promise.all and the job goes from nine minutes to forty seconds, once. The next run returns 403 on two thirds of the requests. You check the quota, because a 403 from GitHub means the quota, and the quota says four thousand eight hundred requests left. Both numbers are true. They are about different limits.

[github_concurrency_probe.py](#)

<https://www.allanninal.dev/github/secondary-limit-concurrency/>

<https://github.com/allanninal/github-api-fixes/tree/main/secondary-limit-concurrency>

29 **bulk issue or comment creation exceeds 80 requests a minute**

The migration script imports 2,400 issues from the old tracker. It gets through about eighty of them, then every remaining call comes back 403. You check the quota and there are 4,900 requests left in it. The limit that stopped you is not counting requests. It is counting the things you created.

[github_content_burst_audit.py](#)

<https://www.allanninal.dev/github/secondary-limit-content-creation/>

<https://github.com/allanninal/github-api-fixes/tree/main/secondary-limit-content-creation>

30 **a hot endpoint burns 900 points a minute and gets throttled**

One endpoint in the job keeps failing and the rest are fine. It fails for about a minute, recovers, and fails again twenty minutes later. The hourly quota is barely touched and the concurrency is one, because someone already serialised it after the last incident. What is left is the cap nobody budgets for: not how many requests you make, but how much work you asked one path to do inside sixty seconds.

[github_endpoint_cost_audit.py](#)

<https://www.allanninal.dev/github/secondary-limit-points-per-minute/>

<https://github.com/allanninal/github-api-fixes/tree/main/secondary-limit-points-per-minute>

31 **the token expires in days and nothing is watching the clock**

Nothing is broken. That is the entire point of this note: it is the one that runs before the outage rather than after it. Somewhere in your environment is a credential with six days left on it, and the only thing standing between you and a 09:14 Tuesday is that nobody has read a header GitHub has been sending on every single response for months.

[github_token_expiry_watch.py](#)

<https://www.allanninal.dev/github/token-expiring-soon/>

<https://github.com/allanninal/github-api-fixes/tree/main/token-expiring-soon>

32 **the token is passed as an access_token query parameter**

The call returns 401 {"message": "Requires authentication"} for an endpoint that plainly ought to work, and the token in the URL is correct — you can paste it into a header and watch the same call succeed. That is the small half of this. The large half is that the URL is now in an access log, a CI transcript, a browser history and a support ticket, and nothing about that half produces an error at all.

[github_token_in_url.py](#)

<https://www.allanninal.dev/github/token-in-query-string/>

<https://github.com/allanninal/github-api-fixes/tree/main/token-in-query-string>

33 **a pinned X-GitHub-API-Version stopped being supported**

Nothing was deployed. No token was rotated, no permission changed, no dependency was upgraded. On a Tuesday morning every request to the GitHub API starts coming back refused, with a message about an API version, and the only thing that moved is the date. The header responsible was copied out of a documentation sample in 2022 and has not been looked at since.

[github_api_version_pin.py](#)

<https://www.allanninal.dev/github/unsupported-api-version/>

<https://github.com/allanninal/github-api-fixes/tree/main/unsupported-api-version>

34 **a classic token nobody used for a year is deleted for you**

The restore drill is on a Tuesday morning and the token that has been sitting in the vault since the runbook was written answers 401. It has not expired, because it was created with no expiry. Nobody revoked it. It is not in the account's token list at all, because GitHub removed it for going a year without being used, and the reason it went a year without being used is that it is a token for emergencies.

[github_token_dormancy.py](#)

<https://www.allanninal.dev/github/unused-classic-token-auto-revoked/>

<https://github.com/allanninal/github-api-fixes/tree/main/unused-classic-token-auto-revoked>

35 **every request 403s because there is no User-Agent header**

403 {"message":"Request forbidden by administrative rules. Please make sure your request has a User-Agent header."} — and it happens on the REST root, which any anonymous caller can read. Nobody looks at the body, because 403 in this API almost always means quota or permissions, and both of those theories send you somewhere the answer is not.

[github_user_agent_403.py](#)

<https://www.allanninal.dev/github/user-agent-missing/>

<https://github.com/allanninal/github-api-fixes/tree/main/user-agent-missing>

36 **webhook deliveries are failing and nobody reads the log**

Someone opens a pull request and the bot that should comment on it says nothing. You grep your receiver's logs for the last hour and find no request at all, which points the finger at GitHub. It is not GitHub. The delivery happened, your server answered 502, and GitHub wrote that down in a log you have never opened.

[github_hook_delivery_audit.py](#)

<https://www.allanninal.dev/github/webhook-deliveries-failing/>

<https://github.com/allanninal/github-api-fixes/tree/main/webhook-deliveries-failing>

37 **the hook is not subscribed to the event you are waiting for**

The handler was written, reviewed, unit tested against a saved payload, and deployed. In production it has never executed once. There is no error anywhere because there is no failure anywhere: the hook was created years ago with push and pull_request, and the event your handler waits for has never been sent to it.

[github_hook_event_coverage.py](#)

<https://www.allanninal.dev/github/webhook-event-not-subscribed/>

<https://github.com/allanninal/github-api-fixes/tree/main/webhook-event-not-subscribed>

38 **a webhook with no secret sends no signature to verify**

The receiver has a signature check. It reads X-Hub-Signature-256, computes an HMAC over the body, compares in constant time, and returns 401 when they differ. It has never returned 401, because the hook it serves has no secret, so GitHub does not send the header, and the check quietly skips itself on every request.

[github_hook_secret_audit.py](#)

<https://www.allanninal.dev/github/webhook-no-secret/>

<https://github.com/allanninal/github-api-fixes/tree/main/webhook-no-secret>

39 **a JWT sent as token, and the 401 blames the credential**

The App is registered, the private key is the right one, the JWT is freshly signed and it verifies in three different debuggers. GitHub answers 401 {"message":"Bad credentials"}. The credential in that request is perfect. The word in front of it is not, and the message never mentions words.

[github_auth_scheme.py](#)

<https://www.allanninal.dev/github/wrong-authorization-scheme/>

<https://github.com/allanninal/github-api-fixes/tree/main/wrong-authorization-scheme>

40 **the automation runs as a person who can leave the company**

The release notes are signed by someone who left in March. Every automated review comment carries a colleague's avatar, every bot commit says their name, and none of it is a display problem: the token doing the work was minted on their account four years ago, so the integration is not running as a service. It is running as them.

[github_actor_identity.py](#)

<https://www.allanninal.dev/github/wrong-identity-token/>

<https://github.com/allanninal/github-api-fixes/tree/main/wrong-identity-token>
